Welcome

COURSE:
ADVANCED PROJECT MANAGEMENT IN AI

Lesson 3 Session 1

Resource Management in AI Projects:
Human Resource Management

Welcome

课程：
人工智能高级项目管理

第 3 课 第 1 节

人工智能项目的资源管理：
人力资源管理

LESSON 3.1 – Human Resource Management in AI Projects



第3.1课：人工智能项目的人力资源管理



2. Learning Objectives: Lesson 3.1

- Explain the role of project managers and business stakeholders in AI projects
- Identify key project constraints in AI initiatives, including scope, time, cost, and quality
- Describe the structure and responsibilities of core roles in AI project teams
- Understand how effective collaboration between business and technical teams supports successful AI implementation

2. 第3.1课的学习目标

- 阐述项目经理与业务相关方在人工智能项目中的角色
- 识别人工智能项目的关键制约因素，包括范围、时间、成本和质量
- 描述人工智能项目团队核心角色的架构与职责
- 理解业务团队与技术团队的有效协作如何助力人工智能项目的成功落地

Human Resource Management in AI Projects

- AI initiatives require collaboration between business leaders and technical specialists
- Projects involve multidisciplinary expertise
- Effective coordination is essential for translating business needs into AI solutions
- Project managers play a central role in integrating business and technical perspectives

人工智能项目的人力资源管理

- 人工智能项目需要业务负责人与技术专家协作开展
- 涉及多领域专业知识
- 有效的协调工作是将业务需求转化为人工智能解决方案的关键
- 项目经理在整合业务与技术视角方面发挥核心作用。

The Strategic Role of Project Managers in AI Projects

Project Management Responsibilities

- Translate business problems into structured project objectives
- Coordinate collaboration across technical and business teams
- Manage experimentation and uncertainty in model development
- Align technical outcomes with organizational strategy
- Oversee project constraints: scope, time, cost, and quality

AI project managers guide iterative experimentation processes rather than purely deterministic development cycles.

项目经理在人工智能项目中的战略角色

项目管理职责

- 将业务问题转化为结构化的项目目标
- 协调技术团队与业务团队间的协作
- 管理模型开发过程中的实验工作与不确定性因素
- 使技术成果与组织战略保持一致
- 把控项目的各项制约因素：范围、时间、成本和质量

人工智能项目经理主导的是迭代式的实验流程，而非纯粹确定性的开发周期。

Defining the Business Problem Clearly

Importance of Problem Definition

- Identify the specific decision the AI system should support
- Define measurable business outcomes
- Avoid vague objectives such as “use AI to improve operations”
- Translate business problems into analytical tasks such as prediction or classification

Clear problem definition improves data selection, model design, and evaluation criteria.

Example:
Instead of: “Use AI to improve customer service”
Define: “Predict the probability of customer churn within 30 days.”

清晰界定业务问题

问题界定的重要性

- 明确人工智能系统需支持的具体决策
- 设定可量化的业务成果
- 避免“运用人工智能优化运营”这类模糊的目标
- 将业务问题转化为预测、分类等分析任务

示例
模糊表述：“运用人工智能提升客户服务质量”
精准界定：“预测客户 30 天内的流失概率”

清晰的问题界定有助于优化数据选择、模型设计与评估标准。

Managing Project Constraints in AI Projects

Core Project Constraints

Project managers must balance:

- Scope – clearly define what the AI system will and will not deliver
- Time – plan for experimentation cycles and model iteration
- Cost – account for infrastructure, talent, and data acquisition
- Quality – define acceptable model performance metrics

AI projects require flexibility because performance outcomes are uncertain during early development stages.

管理人工智能项目的制约因素

核心项目制约因素

项目经理需平衡以下因素：

- 范围 —— 明确界定人工智能系统的交付边界
- 时间 —— 为实验周期和模型迭代制定规划
- 成本 —— 核算基础设施、人才与数据获取的相关投入
- 质量 —— 设定可接受的模型性能指标

人工智能项目的开发初期存在性能结果的不确定性，因此需要保持灵活性。

Role of Business Stakeholders

Business Stakeholder Contributions

- Provide operational context and domain understanding
- Define decision processes supported by the AI system
- Identify regulatory or organizational constraints
- Evaluate whether model outputs are practically useful

AI models may be technically accurate but operationally irrelevant without business validation.

业务相关方的角色

业务相关方的工作贡献

- 提供业务运营背景与领域专业认知
- 明确人工智能系统所支持的决策流程
- 识别相关的监管与组织制约条件
- 评估模型输出结果的实际应用价值

缺乏业务层面的验证，即便人工智能模型在技术层面精准，也可能不具备实际运营价值。

Contract and Vendor Management in AI Projects

Possible External Partners in AI Initiatives:

- AI consulting firms
- Cloud service providers
- Data providers
- Specialized development partners

Clear contracts reduce legal, operational, and financial risks.

Project managers must clarify:

- Data ownership and usage rights
- Intellectual property for models and algorithms
- Service-level expectations
- Long-term maintenance responsibilities

人工智能项目的合同与供应商管理

人工智能项目的潜在外部合作方：

- 人工智能咨询公司
- 云服务提供商
- 数据供应商
- 专业开发合作方

清晰的合同能降低法律、运营与财务层面的风险。

项目经理需明确以下事项：

- 数据的所有权与使用权
- 模型和算法的知识产权归属
- 服务水平预期
- 长期的维护责任

Structure of an AI Project Team

Typical AI Project Team Structure

AI projects require collaboration among several specialized roles:

- Data scientists
- Machine learning engineers
- Data engineers
- Domain experts
- Product owners or project managers

These professionals collectively support data preparation, model development, deployment, and operational integration.

人工智能项目团队的架构

典型的人工智能项目团队架构

人工智能项目需要多个专业角色协作开展：

- 数据科学家
- 机器学习工程师
- 数据工程师
- 领域专家
- 产品负责人或项目经理

项目团队成员协同完成数据准备、模型开发、部署及业务运营整合工作。

Role of Data Scientists

Primary Responsibilities

- Analyze datasets to identify patterns and relationships
- Develop and train machine learning models
- Select appropriate algorithms and modeling approaches
- Evaluate model performance using statistical metrics

Data scientists focus on developing predictive models from data.

Core Skills

- Statistics and machine learning
- Programming languages such as Python or R
- Data analysis and visualization

数据科学家的角色

主要职责

- 分析数据集，识别数据规律与关联关系
- 开发并训练机器学习模型
- 选择适用的算法与建模方法
- 运用统计指标评估模型性能

数据科学家的核心工作是从数据中开发预测模型。

核心技能

- 统计学与机器学习知识
- Python、R 等编程语言
- 数据分析与可视化能力

Role of Machine Learning Engineers

Primary Responsibilities

- Deploy machine learning models into production environments
- Optimize models for scalability and performance
- Build APIs or services for real-time predictions
- Maintain automated pipelines for model updates

Core Skills

- Software engineering
- Distributed systems
- Model optimization and infrastructure management

ML engineers ensure models operate reliably within operational systems.

机器学习工程师的角色

主要职责

- 将机器学习模型部署至生产环境
- 优化模型的可扩展性与性能
- 搭建支持实时预测的应用程序接口或服务
- 构建模型更新的自动化流水线

核心技能

- 软件工程能力
- 分布式系统知识
- 模型优化与基础设施管理能力

机器学习工程师确保模型在业务系统中稳定运行。

Role of Data Engineers

Primary Responsibilities

- Design data pipelines and data infrastructure
- Collect, clean, and transform large datasets
- Manage data storage systems
- Ensure reliable access to data for model training and inference

Core Skills

- Database technologies
- Data pipeline development
- Cloud data platforms; Big data processing frameworks

Data engineers ensure consistent and high-quality data availability.

数据工程师的角色

主要职责

- 设计数据流水线与数据基础设施
- 收集、清洗并转换大规模数据集
- 管理数据存储系统
- 保障模型训练与推理过程的稳定数据获取

核心技能

- 数据库技术
- 数据流水线开发能力
- 云数据平台与大数据处理框架的应用能力

数据工程师保障数据的持续高质量供给。

Role of Domain Experts

Importance of Domain Expertise

Domain experts provide:

- Deep knowledge of business processes
- Identification of meaningful variables for modeling
- Interpretation of model outputs within operational contexts
- Validation of AI recommendations

Examples:

- Physicians in healthcare diagnostics
- Financial analysts in fraud detection
- Engineers in predictive maintenance systems

领域专家的角色

领域专业知识的重要性

领域专家能够：

- 提供深入的业务流程认知
- 识别建模所需的关键变量
- 结合业务场景解读模型输出结果
- 验证人工智能建议的合理性

示例

- 医疗诊断领域的医生
- 反欺诈领域的金融分析师
- 预测性维护系统领域的工程师

Key Takeaways

Human Resource Management Insights

- AI projects require strong collaboration between business leaders and technical specialists
- Clear problem definition improves alignment between technical work and business outcomes
- Project managers must manage constraints, stakeholders, and external vendors
- Specialized roles such as data scientists, ML engineers, and data engineers enable technical implementation

Successful AI initiatives depend on effective coordination across all these roles.

核心要点

人力资源管理启示

- 人工智能项目需要业务负责人与技术专家的紧密协作
- 清晰的问题界定能提升技术工作与业务成果的契合度
- 项目经理需把控项目制约因素、协调相关方并管理外部供应商
- 数据科学家、机器学习工程师、数据工程师等专业角色是技术落地的关键

人工智能项目的成功，依赖于所有相关角色的高效协作。

THANKS !!!



谢谢！



Welcome

COURSE:
ADVANCED PROJECT MANAGEMENT IN AI

Lesson 3 Session 2

Resource Management in AI Projects:
Technical Infrastructure and Tools

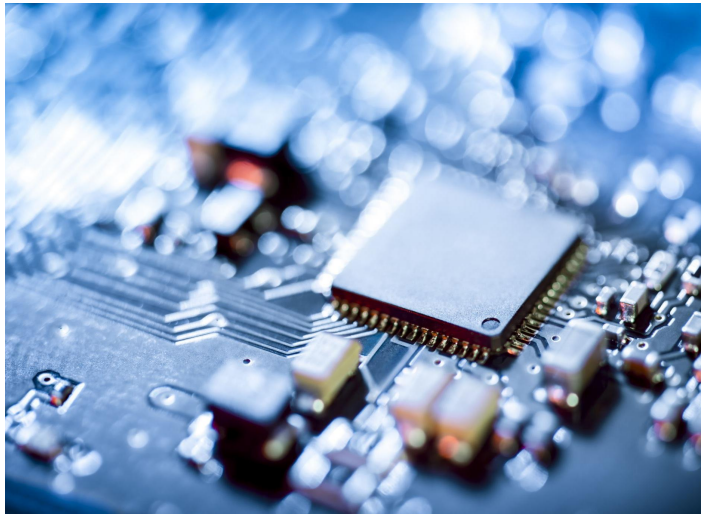
Welcome

课程：
人工智能高级项目管理

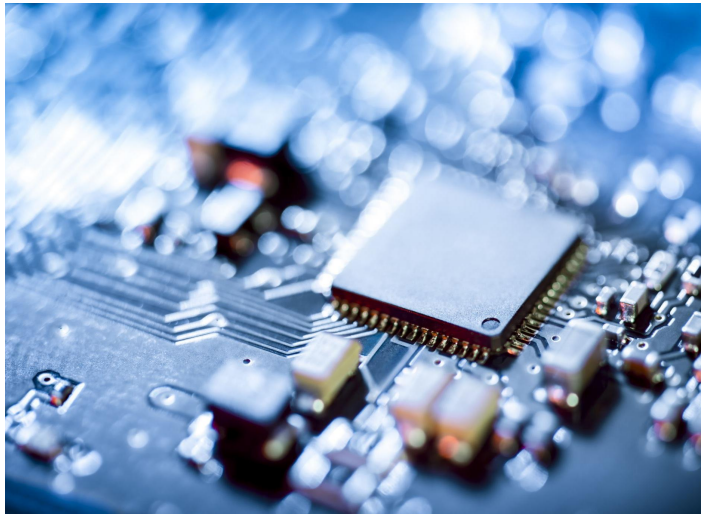
第 3 课 第 2 节

人工智能项目的资源管理：
技术基础设施与工具

LESSON 3.2 – Technical Infrastructure and Tools



第3.2课：技术基础设施与工具



Lesson 3.2 – Technical Infrastructure and Tools

Learning Objectives

- Identify the key technical infrastructure components required in AI projects
- Explain the role of cloud platforms, compute resources, and data pipelines in AI development
- Understand how MLOps tools support model lifecycle management
- Evaluate infrastructure considerations that influence AI project planning and resource decisions

第3.2课 技术基础设施与工具

学习目标

- 识别人工智能项目所需的关键技术基础设施组件
- 阐述云平台、计算资源与数据流水线在人工智能开发中的作用
- 理解机器学习运维工具对模型生命周期管理的支撑作用
- 分析影响人工智能项目规划与资源决策的基础设施考量因素

What Project Managers Need to Know About AI Infrastructure

Project managers are not expected to configure AI infrastructure, but they must understand its implications for project planning and execution.

PMs should understand:

- how infrastructure affects project timelines and experimentation cycles
- the resource requirements for training and deploying AI models
- dependencies between data availability, compute resources, and development tools
- the cost and scalability implications of infrastructure choices

This understanding allows PMs to coordinate technical teams effectively, anticipate risks, and make informed planning decisions

(Sculley et al., 2015; PMI, 2021).

项目经理需掌握的人工智能基础设施知识

项目经理无需配置人工智能基础设施，但必须理解其对项目规划与执行的影响。

项目经理应掌握以下内容：

- 基础设施如何影响项目工期与实验周期
- 人工智能模型训练与部署的资源需求
- 数据可用性、计算资源与开发工具之间的依赖关系
- 基础设施选择对成本及可扩展性的影响

掌握这些内容，项目经理才能有效协调技术团队、预判风险，并做出科学的规划决策。

(Sculley et al., 2015; PMI, 2021).

Introduction to AI Infrastructure

- Infrastructure acts as a foundation for AI development and deployment
- AI projects require different technical environments compared to traditional software systems
- AI workflows involve large datasets, complex models, and iterative experimentation
- Effective infrastructure enables teams to move from prototype to production systems

AI infrastructure supports the entire lifecycle of machine learning systems

(Zaharia et al., 2018).

人工智能基础设施概述

- 基础设施是人工智能开发与部署的基础
- 人工智能项目所需的技术环境与传统软件系统存在差异
- 人工智能工作流程涉及大规模数据集、复杂模型与迭代式实验
- 完善的基础设施助力团队实现从原型到生产系统的落地

人工智能基础设施为机器学习系统的全生命周期提供支撑。

(Zaharia et al., 2018).

Why Infrastructure Matters in AI Projects

- AI systems are computationally intensive
- Large volumes of structured and unstructured data must be processed
- Model training requires specialized hardware and software environments
- Infrastructure must support experimentation, iteration, and rapid scaling

Infrastructure determines how efficiently teams can develop, test, and deploy AI models

(Amershi et al., 2019).

基础设施在人工智能项目中的重要性

- 人工智能系统的计算量较大
- 需处理海量的结构化与非结构化数据
- 模型训练需要专用的软硬件环境
- 基础设施需支持实验、迭代与快速扩展

基础设施决定了团队开发、测试与部署人工智能模型的效率。

(Amershi et al., 2019).

Core Infrastructure Components in AI Projects

Typical AI infrastructure stack includes:

- data storage and data pipelines
- compute resources (CPU, GPU, distributed systems)
- machine learning frameworks
- development environments
- model deployment platforms
- monitoring and lifecycle management tools

These components collectively support the full lifecycle of AI systems

(Sculley et al., 2015).

人工智能项目的核心基础设施组件

典型的人工智能基础设施体系包含：

- 数据存储与数据流水线
- 计算资源（中央处理器、图形处理器、分布式系统）
- 机器学习框架
- 开发环境
- 模型部署平台
- 监控与生命周期管理工具

这些组件共同支撑人工智能系统的全生命周期运营。

(Sculley et al., 2015).

Cloud Platforms for AI Development

Major cloud platforms supporting AI projects include:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)

Key capabilities include:

- scalable compute resources
- managed machine learning services
- integrated data storage solutions
- prebuilt AI tools and APIs

Cloud platforms allow organizations to scale infrastructure on demand and reduce capital investment

人工智能开发的云平台

主流人工智能项目云平台：

- 亚马逊网络服务
- 微软 azure 云平台
- 谷歌云平台

核心功能：

- 可扩展的计算资源
- 托管式机器学习服务
- 集成化数据存储解决方案
- 预制人工智能工具与应用程序接口

云平台支持企业按需扩展基础设施，降低资本投入。

Compute Infrastructure: CPUs and GPUs

AI model development often requires specialized computing hardware.

CPUs

- general-purpose processors
- suitable for data processing and smaller models

GPUs

- parallel processing architecture
- highly effective for training deep neural networks

Modern deep learning systems rely heavily on GPU acceleration to handle large-scale computations

计算基础设施：中央处理器与图形处理器

人工智能模型开发通常需要专用的计算硬件。

中央处理器

- 通用处理器
- 适用于数据处理与小型模型开发

图形处理器

- 采用并行处理架构
- 在深度神经网络训练中表现优异

现代深度学习系统的大规模计算高度依赖图形处理器的加速能力。

Data Infrastructure and Pipelines

Reliable data pipelines are essential for AI systems.

Key elements include:

- data ingestion from internal and external sources
- data cleaning and preprocessing
- feature engineering processes
- storage systems such as data lakes or warehouses
- continuous data updates for retraining models

Poor data management can significantly hinder AI project success (Amershi et al., 2019).

数据基础设施与数据流水线

可靠的数据流水线是人工智能系统的核心要素，其关键组成包括：

- 从内外部数据源采集数据
- 数据清洗与预处理
- 特征工程处理
- 数据湖、数据仓库等存储系统
- 为模型再训练提供持续的数据更新

数据管理不当会严重阻碍人工智能项目的推进。

(Amershi et al., 2019).

Machine Learning Frameworks and Development Tools

Common machine learning frameworks include:

- TensorFlow
- PyTorch
- Scikit-learn

These frameworks support:

- model development and training
- algorithm implementation
- experimentation and evaluation
- integration with deployment platforms

Standardized frameworks enable faster experimentation and collaboration across teams (Zaharia et al., 2018).

机器学习框架与开发工具

常用机器学习框架

- 张量流
- 火炬框架
- 思凯乐学习库

框架核心功能

- 模型开发与训练
- 算法实现
- 实验与评估
- 与部署平台的集成

标准化框架能加快实验进程，促进团队间的协作。

(Zaharia et al., 2018).

MLOps and Lifecycle Management

MLOps (Machine Learning Operations) extends DevOps principles to machine learning systems.

Key capabilities include:

- automated model training pipelines
- model deployment workflows
- experiment tracking
- performance monitoring
- retraining and updating models

MLOps enables organizations to manage AI systems continuously throughout their lifecycle

机器学习运维与生命周期管理

机器学习运维将开发运维的原则延伸至机器学习系统。

其核心功能包括：

- 模型训练自动化流水线
- 模型部署工作流
- 实验跟踪
- 性能监控
- 模型再训练与更新

机器学习运维助力企业对人工智能系统进行全生命周期的持续管理。

Model Version Control and Reproducibility

AI models evolve continuously during development.

Projects must track:

- model versions
- datasets used for training
- parameter configurations
- evaluation metrics

Version control tools help ensure reproducibility, traceability, and governance of machine learning systems

(Sculley et al., 2015).

模型版本控制与可复现性

人工智能模型在开发过程中持续迭代。

项目需对以下内容进行跟踪：

- 模型版本
- 训练所用数据集
- 参数配置
- 评估指标

版本控制工具保障机器学习系统的可复现性、可追溯性与规范化管理。

(Sculley et al., 2015).

Infrastructure Challenges in AI Projects

Organizations commonly face infrastructure challenges such as:

- high compute costs for training large models
- managing large datasets securely
- integrating AI systems with existing IT infrastructure
- maintaining consistent development environments across teams

Infrastructure complexity often creates operational challenges in scaling AI systems (Amershi et al., 2019).

人工智能项目的基础设施挑战

企业在基础设施方面常面临以下挑战：

- 大型模型训练的计算成本高昂
- 大规模数据集的安全管理
- 人工智能系统与现有信息技术基础设施的集成
- 保障跨团队开发环境的一致性

基础设施的复杂性往往会给人工智能系统的规模化落地带来运营挑战。

(Amershi et al., 2019).

Key Takeaways

From a project management perspective, infrastructure affects:

- budget allocation and cost forecasting
- project scheduling and iteration cycles
- coordination between technical and business teams
- long-term scalability of AI solutions

Even without deep technical expertise, project managers must understand these elements to support governance and decision-making in AI initiatives.

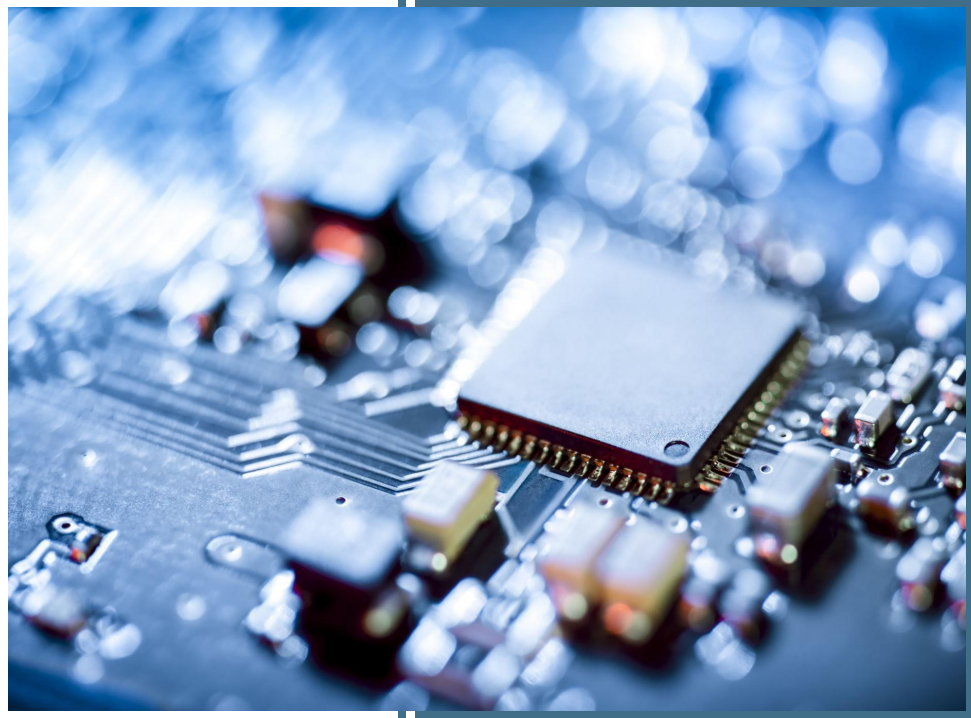
核心要点

从项目管理视角来看，基础设施对以下方面产生影响：

- 预算分配与成本预测
- 项目进度规划与迭代周期
- 技术团队与业务团队的协作
- 人工智能解决方案的长期可扩展性

即便不具备深厚的技术专业知识，项目经理也必须掌握这些内容，才能为人工智能项目的治理与决策提供支撑。

THANKS !!!



谢谢！



Welcome

COURSE:
ADVANCED PROJECT MANAGEMENT IN AI

Lesson 3 Session 3

Resource Management in AI Projects:
Budgeting and Cost Estimation

Welcome

课程：
人工智能高级项目管理

第 3 课 第 3 节

人工智能项目的资源管理：
预算编制与成本估算

LESSON 3.3 – Budgeting and Cost Estimation in AI Projects



第 3.3 课 —— 人工智能项目的预算编制与成本估算



Learning Objectives

- Identify the major cost components in AI projects
- Estimate infrastructure, compute, and personnel costs
- Understand cost uncertainty in experimental AI development
- Apply structured budgeting approaches for AI initiatives

学习目标

- 识别人工智能项目的主要成本构成
- 估算基础设施、计算与人力成本
- 理解实验性人工智能开发中的成本不确定性
- 为人工智能项目运用结构化的预算编制方法

Why Budgeting Matters in AI Projects (PM Perspective)

Budgeting Challenges in AI Projects

- AI projects involve highly variable cost structures
- Data, compute, and experimentation create unpredictable spending patterns
- Traditional IT estimation models often underestimate AI costs
- Project managers must balance innovation with financial accountability
- Effective cost estimation improves stakeholder trust and project viability

预算编制在人工智能项目中的重要性（项目经理视角）

人工智能项目的预算编制挑战

- 成本结构的可变性高
- 数据、计算与实验工作导致支出模式难以预测
- 传统信息技术成本估算模型往往低估人工智能项目成本
- 项目经理需在创新与财务管控间寻求平衡
- 科学的成本估算能提升相关方的信任度与项目的可行性

Major Cost Categories in AI Projects

Core Cost Components

AI projects typically involve five major cost areas:

- Data acquisition and preparation
- Infrastructure and storage
- Compute and model training resources
- Personnel and specialized talent
- Deployment and operational costs

These costs interact and evolve throughout the AI lifecycle.

人工智能项目的主要成本类别

核心成本构成

人工智能项目的成本主要分为五大类：

- 数据获取与准备成本
- 基础设施与存储成本
- 计算与模型训练资源成本
- 人力与专业人才成本
- 部署与运营成本

这些成本在人工智能项目的全生命周期中相互影响、动态变化。

Data Acquisition and Preparation Costs

Data as a Primary Cost Driver

Key cost elements include:

- Purchasing external datasets
- Data collection (sensors, surveys, logs)
- Data labeling and annotation
- Data cleaning and preprocessing
- Data governance and compliance

Data preparation can account for a large portion of project effort and cost.

数据获取与准备成本

数据作为核心成本驱动因素

关键成本要素包括：

- 外购数据集
- 数据收集（传感器、调研、日志）
- 数据标注
- 数据清洗与预处理
- 数据治理与合规相关工作

数据准备工作往往占据项目大量的人力与成本投入。

Infrastructure and Storage Costs

Infrastructure Requirements

Typical infrastructure costs include:

- Cloud storage for large datasets
- Data pipelines and ETL systems
- Distributed computing environments
- Database and data warehouse services
- Security and data management systems

Organizations must decide between cloud, on-premise, or hybrid infrastructure models.

基础设施与存储成本

基础设施需求

典型的基础设施成本包括：

- 大规模数据集的云存储费用
- 数据流水线与抽取、转换、加载系统成本
- 分布式计算环境搭建成本
- 数据库与数据仓库服务费用
- 安全与数据管理系统成本

企业需在云部署、本地部署与混合部署的基础设施模式中做出选择。

Compute and Model Training Costs

Computational Resource Costs

Important cost drivers include:

- GPU or TPU computing resources
- Training time and experimentation cycles
- Hyperparameter tuning processes
- Model retraining during development
- Scaling resources for large datasets

Advanced models such as deep learning systems can require significant computational investment.

计算与模型训练成本

计算资源成本

主要成本驱动因素包括：

- 图形处理器、张量处理单元等计算资源投入
- 训练时长与实验周期
- 超参数调优过程
- 开发过程中的模型再训练
- 为大规模数据集调配的资源扩展成本

深度学习等先进模型的开发需要大量的计算投入。

Personnel and Talent Costs

Human Resource Costs in AI Projects

AI initiatives require specialized expertise.

Key roles include:

- Data scientists
- Machine learning engineers
- Data engineers
- Domain experts
- Project managers and product owners

Due to talent scarcity, personnel costs often represent a major portion of AI project budgets.

人力与专业人才成本

人工智能项目的人力资源成本

人工智能项目需要专业的技术人才。

核心岗位包括：

- 数据科学家
- 机器学习工程师
- 数据工程师
- 领域专家
- 项目经理与产品负责人

由于专业人才稀缺，人力成本通常是人工智能项目预算的重要组成部分。

Deployment and Operational Costs

Costs Beyond Model Development

Operational costs include:

- Production deployment infrastructure
- Monitoring and performance tracking
- Model retraining and updates
- Integration with existing systems
- User training and adoption support

Successful AI systems require ongoing operational investment.

部署与运营成本

模型开发之外的成本

运营成本包括：

- 生产环境部署的基础设施成本
- 监控与性能跟踪成本
- 模型再训练与更新成本
- 与现有系统的集成成本
- 用户培训与落地支持成本

成功的人工智能系统需要持续的运营投入。

Cost Uncertainty in AI Projects

Sources of Budget Uncertainty

AI development introduces several uncertainties:

- Data availability and quality issues
- Model performance variability
- Experimentation cycles and failed models
- Changing business requirements
- Regulatory or compliance constraints

Budgeting therefore requires flexibility and contingency planning.

人工智能项目成本的不确定性

预算不确定性的来源

人工智能开发过程中存在多种不确定性因素：

- 数据的可用性与质量问题
- 模型性能的波动
- 实验周期与模型开发失败的成本
- 不断变化的业务需求
- 监管与合规约束的调整

因此，预算编制需要保持灵活性并制定应急方案。

Cost Estimation Techniques for AI Projects

Approaches for Estimating AI Costs

Project managers can apply:

- Analogous estimation using past projects
- Bottom-up estimation from task-level activities
- Parametric estimation based on resource usage
- Scenario-based estimation for experimentation stages

Combining techniques improves estimation reliability.

人工智能项目的成本估算方法

人工智能成本估算方式

项目经理可采用以下方法：

- 类比估算：参考过往项目数据
- 自下而上估算：基于具体任务活动核算
- 参数估算：依据资源使用情况计算
- 情景估算：针对实验阶段制定不同情景的预算

多种方法结合使用能提升估算的准确性。

Budget Buffers and Contingency Planning

Managing Financial Risk

Best practices include:

- Allocating experimentation buffers
- Creating phased funding milestones
- Monitoring compute and infrastructure costs
- Implementing cost tracking dashboards
- Reviewing budgets at key project checkpoints

These practices help control cost escalation while enabling experimentation.

预算缓冲与应急规划

财务风险管理的最佳实践

- 为实验工作预留预算缓冲资金
- 设定分阶段的资金拨付节点
- 监控计算与基础设施成本
- 搭建成本跟踪仪表盘
- 在项目关键节点开展预算审核

这些做法能在控制成本攀升的同时，为实验工作提供支持。

Strategic Cost Management in AI Projects

Balancing Innovation and Financial Control

Project managers should focus on:

- Aligning AI investments with business value
- Prioritizing high-impact use cases
- Evaluating cost–benefit trade-offs
- Scaling infrastructure gradually
- Monitoring financial performance continuously

Strategic cost management ensures sustainable AI adoption.

人工智能项目的战略成本管理

在创新与财务管控间寻求平衡

项目经理应聚焦以下方面：

- 使人工智能投资与业务价值相匹配
- 优先推进高价值应用场景
- 评估成本与收益的权衡关系
- 逐步扩展基础设施规模
- 持续监控财务绩效

战略成本管理助力企业实现人工智能的可持续落地。

Key Takeaways

- AI projects have unique cost structures driven by data, compute, and specialized talent
- Budgeting must account for both development and operational phases
- Cost estimation is challenging due to experimentation and model uncertainty
- Project managers should apply structured estimation methods and contingency buffers
- Strategic financial planning helps organizations balance innovation with economic viability

核心要点

- 数据、计算与专业人才是人工智能项目成本结构的核心驱动因素
- 预算编制需同时覆盖开发阶段与运营阶段
- 实验工作与模型的不确定性增加了成本估算的难度
- 项目经理应采用结构化的估算方法并设置预算缓冲
- 科学的财务规划帮助企业在创新与经济可行性间实现平衡

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THANKS !!!



谢谢！

